PRIORITY QUEUES

Problem Solving with Computers-II





Emergency Room Triage

You're a software engineer at a hospital.
Patients arrive and need to be treated in order of severity.

1 = Critical (life-threatening)
2 = Urgent
3 = Moderate
4 = Minor

Emergency Room Triage

What if we used std::queue?

```
    t=1  Alice   (Moderate)   ← treated first
    t=2  Bob     (Critical)   ← waits behind Alice
    t=3  Carol   (Minor)
    ...
```

First come, first served — is that okay here?

C++ Priority Queue ADT

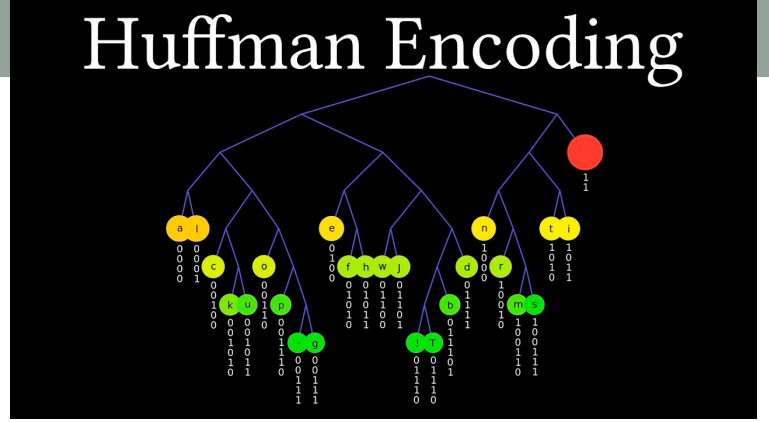
```
priority_queue<int> pq;

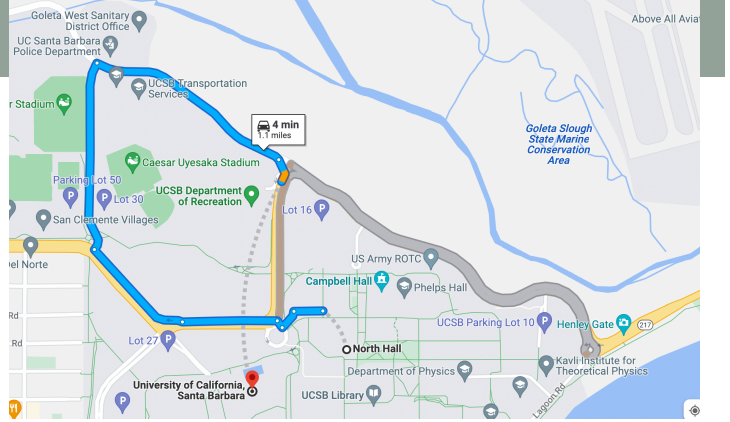
pq.push(20);
pq.push(20);
pq.push(80);
pq.push(50);
pq.push(100);

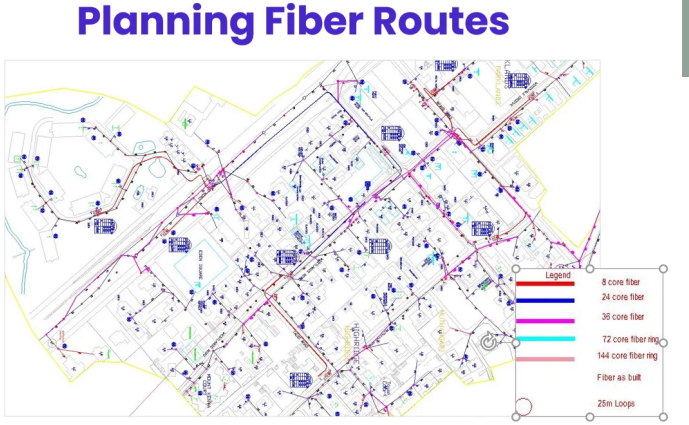
cout << pq.top();

pq.pop();
```

Live code ERTriage







Data Compression | Google Maps Navigation | Network Design

Algorithms: **Huffman Coding** | **Shortest Path** | **Minimum Spanning Tree**

ADT: **Priority Queue**

Data structure: **Binary Heap**

Complete Binary Tree

Vector

Many algorithms need to compute the min OR max repeatedly.

Priority Queue is used to speed up the running time!